

Empirical Study of Hierarchical Intrusion Detection Systems for Unknown Attacks

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Abstract—The attack detection models of the traditional Intrusion Detection Systems (IDSs) are trained on closed-set problems which reduces the classifiers' performance on detecting unknown attacks in the open-set problem space. Mostly adapted, one-shot learning in the classifier does not allow the traditional IDS to be an open-set recognizer. The alternate continuous learning-based IDS in unknown attack detection claims ongoing suggestions from experts to retrain the model with newly identified samples. Hence, using the multi-layer hierarchical IDS (HIDS) with optimized classifier models, the unknown attacks can be classified by comparing their patterns with benign and known attacks. However, we have identified many challenges in the existing HIDS system on various datasets though it provides a solid foundation in this design category for unknown attack identification. As a result, in this paper, we designed an enhanced multi-tier IDS for zero-day attack detection with optimized heterogeneous classifiers in its major two phases like basic framework demands. We have examined the enhanced proposed hierarchical IDS on various benchmark Intrusion Detection Systems datasets such as WUSTL, CIC_IDS_2017, 5G and UNR to analyze the efficiency in unknown attacks classification. When compare to existing multi-tier IDS, the proposed IDS achieved highest detection 96.2%, 87%, 96.8% and 100% in 5G, WUSTL, UNR and CIC_IDS_2017 datasets for unknown attacks. The optimized model in the proposed IDS reduces the time complexity into 50% than the existing. Implementation results show the proposed enhanced IDS performs better than the existing hierarchical IDS with a high true positive rate for benign, known and unknown attack labels on various datasets.

Index Terms—Intrusion detection system, zero-day attack, open set problems, machine learning.

I. INTRODUCTION

THE SCOPE of zero-day attacks in the latest networks such as Internet of Things (IoT), Industrial-IoT (IIoT) and

5G is very high due to their diversified connected devices, vast coverage, large network scalability, and restricted computation resources in the connected devices. Due to advanced intelligent technology [1], openly accessible high-end resources, and the motivations from terrorists, attackers trigger more attacking strategies than before. The zero-day attacks can be predicted only after a while from the attack initiation to create an explicit impact on the network performances. Meanwhile, its impact on the network is substantial. In this situation, the conventional signature-based IDSs make them unable to protect the networks from unknown attacks with zero knowledge about the new coming attacking strategy. The traditional IDSs have been built by learning the patterns of benign and known attack instances. With limited data sets, fewer minority samples and traffic do not contain modern networks' footprints, the conventional IDSs struggle to identify new attacks and new benign traffic [2]. Also, the unknown attack classification gets complicated and produces high false positive rates in such IDSs because of real-world open-set problems [3].

Traditional Artificial Intelligence-enabled IDSs are mostly designed as a closed-set recognition approach where all class samples are used in the training phase and those class labels alone are expected to occur during testing for prediction [3], [4], [5]. The Open Set Recognition (OSR) approach facilitates the classifier to detect zero-day attack instances in an adaptive and proactive manner [3]. To address the open-set problem, IDS should be designed to classify the newly arrived class test samples that are not seen or used to train the classification model [4]. Further, one-shot training is not sufficient for an intrusion detection model to learn the pattern of the zero-day attacks. The IDS framework with OSR [6], [7] continuously evolves the learning even after training to detect unknown attacks. Unsupervised machine learning algorithms such as clustering and outlier analysis can identify the samples that are deviated from benign patterns. This methodology in the existing works [8], [9] can solve the zero-day attack identification at a baseline level.

To further validate our observations, in this paper, we systematically evaluated the performance of multistage, hierarchical IDS which is proposed by Verkerken et al. [10] against unknown attacks. In their work, the IDS has given 96% balanced accuracy in the overall classification and 87% zero-day attack classification for CIC_IDS_2017 dataset. This multi-tier hierarchical IDS has been designed with supervised and semi-supervised algorithms in two tiers and the results are well-tuned with the set of calculated thresholds

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and hyper-parameter optimization algorithm, i.e., TPE (Tree-structured Parzen Estimator) in the classification model. We have analyzed the general solution capacity of this IDS over unknown attacks on other IDS datasets such as UNR [11], 5G_IDD [12], and WUSTL [13]. Interestingly, this system does not produce impressive results in other datasets. In small datasets like UNR and 5G, the classification accuracy for unknown attacks is notable, yet the misclassification rate for known attacks is still quite high. Known attacks are misclassified as unknown attacks. Whereas this system achieves nearly 80% classification accuracy for unknown attacks in large datasets like CIC_IDS_2017 and WUSTL.

Motivated from this, we extended the framework of [10] with various alternative supervised, unsupervised and semi-supervised machine learning and deep learning algorithms in its two stages and evaluated the outcome under permutation and combination of such algorithms. The proposed OSR classifier is also optimized by Gaussian Process-based Bayesian optimisation (BOGP) technique. The implementation results are extensively analyzed to find the best algorithms for anomaly detection (stage1), multi-class known attack detection, and unknown attack (combined stage) detection in the multitier IDS architecture. Our work observed that the optimization algorithm BOGP performs better than TPE and produces very similar high classification accuracy for benign samples like in state-of-the-art [10]. The known attacks are classified by the proposed enhanced OCR IDS with the highest accuracy than in existing IDS. Interestingly this enhanced multi-tier IDS can classify unknown attacks better for CIC_IDS_2017 and UNR datasets. By considering the overall classification accuracy including all labels; benign, known, and unknown attacks, the proposed enhanced multitier IDS gives similar or better values in four datasets.

Our contributions are as follows:

- 1) In our empirical study, we have reproduced the work done by Verkerken et al. [10] in which the authors have proposed a multi-stage IDS against unknown attacks. After our rigorous evaluation of their work, we have investigated that the performance of the model is limited to certain data types and overall, not effective to detect unknown attacks.
- 2) To bridge this identified gap, we have proposed a multitier hierarchical IDS framework with various alternative supervised, unsupervised and semi-supervised machine learning and deep learning algorithms, in its two stages. We have also used BOGP technique to optimize the classification model. These enhancements produce impressive results in the classification rate of unknown, known, and benign attacks. This is a significant contribution in the design of the proposed hierarchical and hybrid intrusion detection system for open set problem that consists of unknown or zero-day attacks.
- 3) Finally, we have further evaluated and compared the performance of the existing and proposed enhanced multi-tier hierarchical IDS on different network datasets. This gives us insights about a) the important role of hyper-parameter optimization algorithm in the

classification model and b) the way to use the appropriate machine learning, deep learning, deep reinforcement learning, and federated learning algorithms for the design of OCR IDS for a zero-day attack detection.

The remainder of this paper is organized as follows: In the next section, existing works in the intrusion detection system designed for unknown or zero-day attacks are discussed. Section III describes the benchmark network intrusion datasets used in this empirical study. Section IV presents various methods and algorithms adapted in the design of an unknown attack detection system. The experimental results and comparison of the performance of various techniques in the hierarchical IDS are presented in Section V. Finally, Section VI concludes the paper.

II. RELATED WORK

The misclassification rate of unknown attacks in conventional single-layer intrusion detection systems is in the unavoidable range that leads to severe data breaches and network instability. In recent years, many initiatives have been taken against unknown or zero-day attacks to design adaptive intrusion detection systems that automatically tune when faced with new patterns in network traffic [8], [9]. Based on this requirement, existing work designed the hierarchical IDS to classify unknown attacks from known attacks and benign traffic. The multilevel, hybrid IDS has been designed with anomaly and signature-based techniques to separate the learning process as benign and adversarial traffic. In the state-of-the-art work [4], [8], [10], [14], [15], benign network activities are learned well by the semi-supervised or unsupervised algorithms, whereas known attacks are learned by the supervised learners. The taxonomy of the existing IDSs for unknown attacks is given in Fig. 1 and summary of such works is presented in Table I.

Zero-day attacks in the Internet of Vehicles (IoV) caused by vehicles and external networks connected to vehicles have been detected by Yang et al. [14] multilevel hybrid (MTH) intrusion detection system. The normal and known traffic patterns are learned by the signature-based IDS. The unknown attack traffic samples are grouped using anomaly-based IDS. They evaluated the efficiency of the designed MTH using the CAN and CIC_IDS_2017 datasets. The limitation of this work is the less F1 score, i.e., 0.8 obtained for the CIC_IDS_2017 data set. A set of appropriate preprocessing steps has been identified to improve the quality of the data set before training. k-means clustering for data sampling, SMOTE-based balanced dataset, label encoding, and z-score normalization algorithm are in the first phase of the MTH-IDS for preprocessing. For training, only optimal features are selected using IG (Information Gain), FCBF (Fast Correlation Based Feature Selection), and KPCA (Kernel Principal Component Analysis) techniques. Signature-based IDS is implemented by supervised algorithms such as Decision Tree, Random Forest, Extra Tree, and XGBoost for known attacks detection. CL-k mean classifier used to implement anomaly-based IDS to label zero-day attacks. Bayesian Optimization-Tree of Parzen Estimator (BO-TPE) [16], [17], [18] and Bayesian optimization based

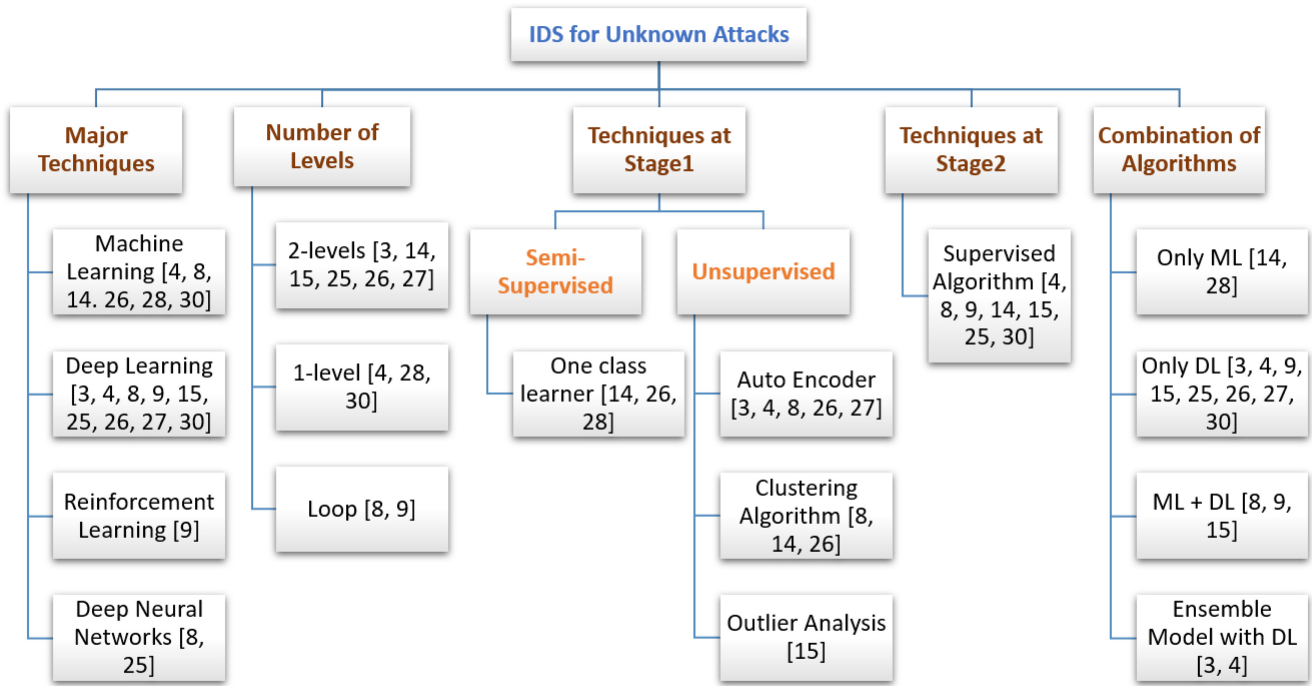


Fig. 1. Taxonomy of Zero-Day Attack IDS.

on Gaussian processes used to optimize the supervised and unsupervised learning algorithms in this MTH.

The model retraining phase is enhanced with Proportional Progressive Pseudo-Labeling (PPPL) technique [19] by excluding uncertain samples in the beginning stages of the training loops. Using the reinforcement learning method [20], [21] or experts labeling, the new class labels are named for the unknown samples and models will be retrained by including those new samples in the original dataset. The performance of IDS on zero-day attacks also improved by the model's hyper-parameters optimization techniques like Bayesian Optimization and Tree-structured Parzen estimator [16]. Also, the zero-day attack signature extraction is very difficult if it overlaps the known attacks or normal traffic samples [22].

In the state-of-the-art work, very limited papers have been published on adaptive IDS to detect zero-day attacks in the open set problem space. The clustering-based open-set recognition models such as DOC (Deep Open Classification), OpenMax, and AutoSVM are initially introduced for zero-day attack detection in the IDS by Soltani et al. [8]. In two ways, the open-set IDS have been designed in state-of-the-work; (i) setting appropriate threshold in the baseline models [10], [23] and (ii) Unsupervised or semi-supervised anomaly detection in the multi-tier IDS framework to filter unknown attacks prior to supervised classification model for known attacks [3], [8], [10].

Soltani et al. [8] designed an adaptable deep learning-based IDS in this category with four phases; (i) Open set recognition used AutoSVM to classify known and unknown attack samples, (ii) unknown attack samples are grouped using a clustering algorithm with deep model in its last layer, (iii) supervised labeling used experts to label clusters of the unknown samples

into a new benign, unknown attack, temporary anomaly and (iv) retrain the model with newly labeled samples to classify zero-day attacks. Another work in this category, proposed by Wu et al. [9] explored deep Q-network (DQN) to design an IDS under active against unknown attacks in the network. Active DQN consists of three phases; (i) NIDS classifier to generate belief vector for benign and malicious samples, (ii) sample selection phase selects valuable samples based on the belief vector and confidence score and (iii) annotator labels the new group samples and added in the existing dataset to retrain the model. With the help of deep learning techniques such as Evaluation Net and Target Net, the deep-Q network reinforcement learning process is enhanced.

Nguyen and Le [15] designed a reliable unknown DoS/DDoS attack detection system for IoT networks using the combination of supervised and unsupervised hybrid IDS. The soft-ordering CNN model is trained by the benign network traffic patterns. The unsupervised algorithms Local Outlier Factor (LOF) and isolated-based Nearest Neighbor Ensemble (iNNE) identified the abnormal traffic patterns in the IoT network. The sequentially deployed SOCNN (Soft Ordering CNN), LOF, and iNNE models detect the test samples as normal or abnormal and forward the abnormal samples to the admin to handle the unknown attacks.

Recently, Fog and Edge Computing Architecture in the IoT network requires edge-device-oriented federated learning to detect zero-day botnet attacks [24]. Further, Popoola et al. [25] applied a Deep Neural Network to create the local and global classification model in different places of the network. The federated learning algorithm aggregates local model updates, and via polling the global model is updated to learn the new samples in the network. Verma et al. [3] designed a zero-day guardian against IIoT in the 5G network using federated

TABLE I
DISCUSSION OF EXISTING IDS FOR UNKNOWN ATTACKS

Title	Type of IDS	Implementation Techniques	Dataset	Model Optimization / Other Optimization	Key Features	Limitations
Zero-day guardian [3]	Multi-level, hierarchical IDS using Federated Learning	FL: Global Autoencoder (AE) Model & One-Class SVM Self Model. Synchronized Duel Models for Normal and Attack	X-IloTID	Polling: Combined Self Learning + Collaborative Learning	Ensemble model of DL and ML. Distributed knowledge sharing for model generalization and scalability	Additional complexity and computation cost
Ensemble DL [4]	Single Level, parallel ensemble model	Ensemble: Autoencoder, CNN & LSTM	CICIDS2017, BoT_IoT, N_BaloT, NSL-KDD	Preprocessing with dimensionality reduction, PCA and ICA	Detailed empirical study has done using 4 datasets	Less impressive result in NSL KDD. Threshold-based detection is not used
Adaptable DL [8]	Multi-level: 4 Phases	(i) Open Set Recognition Phase: DOC++, DOC, OpenMax, and AutoSVM (ii) Clustering (iii) Supervised algorithm (iv) Update the model	CIC-IDS2017	Expert's opinion on classification in the 3rd phase	4 major categories: new normal, new attack, known normal and known attack	human intervention in the decision making as an expert
Active Learning Deep-Q [9]	Loop Multi-level: 3 Phases	(i) Classifier: Benign/Not (ii) Sample selection by DQN and BiLSTM (iii) Annotator: Euclidean distance function	NSL_KDD and UNSW_NB15	Basic Data preprocessing steps	Reinforcement learning algorithm with loop	Computational complexity induced by DQN
MTH-IDS [14]	Single-level, Multi-Tier and Hybrid	Signature based IDS: Stacking ensemble ML models & Anomaly-based IDS: k-means clustering	CAN-intrusion-dataset and CIC-IDS2017	SMOTE & Normalization in data preprocessing. BO-GP optimization for hyper-parameters of the model. IG and FCBF based feature selection	Feature engineering and optimization based hyper-parameters in the models	For the real-time VANET, the important Online learning or adaptive learning is missing
Robust IDS [15]	Hybrid, Multi-level: 3 Phases	Supervised algorithms: SOCNN & Unsupervised algorithm: LOF and iNNE	BoT-IoT, CIC-IDS-2017, and CIC-IDS-2018	Features standardizing	Build resilient model against fast gradient sign method (FGSM) and Carlini Wagner (CW) adversarial attacks	Focuses only on DoS/DDoS (Denial of Service/ Distributed DoS) attacks
FL IDS [25]	FL-based Multi-level	Federated DL with DNN in global and local levels	Bot-IoT and N-BaloT	FedAvg aggregation algorithm for collective learning	General FL benefits like data privacy, low memory etc. LDL, CDL, DDL and FDL performance results are compared	Focuses only on Bot-net attacks. Not integrated the FL optimization techniques
Reliable IDS [26]	Hybrid, Multi-level	Combination of Autoencoder and One-Class classifier for training and profiling using latent representation	Meraz'18 and Drebin	Detection threshold is identified by OC classifier	Zero-day Malware detection system. The proposed model is resilient against synthetically generated adversarials using whitebox (FGSM) and blackbox (HSJA) tools	Improve the model performance against various adversarials like poisoning attacks etc
MS-Zero Wall [27]	Hybrid, Multi-level	VAE-based dual-domain defense strategy (DVAE). k-means anomaly detection and Aggregated Hidden Markov Model	DARPA2000, LLDDOS 1.0 and LLDDOS 2.0.2	AHMM improves the detection rate	Able to find multi-step attacks using multi-step detection	Only 4 multi-step attacks are tested
Semi-Supervised IDS [28]	Single level	Semi-Supervised ML algorithm: One-Class SVM	CICDDoS2019, IOT INTRUSION 2020, CIRA-CIC-DOHBRW-2020 and UNSW-NB15	multivariate Gaussian mixture model (GMM)	Benford's law to find the patterns in the anomaly samples	A specific ML-based IDS
Anomaly-based IDS [30]	Single level	Regularized CNN	BOT-IoT	IG based best feature selection	Can solve over fitting problem using regularisation methods	Model optimization is not considered

learning IDS. Dual synchronized Autoencoders have been used to train the normal and attack samples in this FL-based IDS. Kim et al. [26] used an autoencoder to implement the reliable profiling-based approach to automatically detect zero-day malware. In the training phases, the autoencoder generates the latent representation Z for the training samples, and then Z is passed as an input for the one-class classifier to learn the benign software and malware. One-class learner reads latent representations of the training samples rather than the actual training dataset. The same technique was adopted by Li et al. [27] to design the MS-ZeroWall, a multistep

attack detection system for smart homes. The Variational Autoencoder and Hidden Markov Model are used to learn the known and unknown attack samples. The latent space is very close to the normal Gaussian distribution for known attacks rather than unknown attacks.

Interestingly, Mbona and Eloff [28] and Zoppi et al. [29] used only the semi-supervised machine learning algorithms for zero-day attack detection in the IoT network. Gaussian Mixture Model and the One-Class SVM semi-supervised algorithms have been adapted to design the IDS. In addition to the primary target of detecting zero-day attacks, the IDS is also

designed to optimize its performance in detection accuracy. High-biased and low-variance issues in the prediction of unknown attacks in the model have been reduced by [4] concatenating the results of auto-encoder, CNN, and LSTM in the ensemble Intrusion Detection Systems for IoT and wired networks. The overfitting problem in the training process of unknown attacks IDS was solved by Hairab et al. [30] using CNN along with regularization techniques L1 and L2.

Remarks: The existing works in the open-set IDS design for the unknown or zero-day attack detection are very limited [3], [4], [5], [8], [10]. Also, the IDSs in the state-of-the-art have been designed for a particular network like 5G, IIoT, and IoT. The datasets used in such works are related to specific networks. There is no such general Open-Set IDS for unknown attack classification for different types of networks. In our work, we analyze the performance of out-performed multi-tier hierarchical IDS on unknown attacks that registered on CIC_IDS_2017 dataset against various IDS datasets of different networks. The poor results obtained in our empirical study drive us to enhance the existing hierarchical IDS for zero-day attack classification.

III. DATASETS DESCRIPTION

We have used 4 datasets in our comparative study. The description of the data sets are given below.

CIC_IDS_2017: This dataset is generated by the Canadian Institute of Cybersecurity (CIC) in 2017 [31]. This data set is intended to evaluate the performance of IDS and IPS. This dataset consists of 53,92,441 samples including benign and various attacks such as DDoS, Port Scan, Brute Force, Web-Attack, Botnet, Infiltration, and Heartbleed. For the implementation of the purpose of the empirical study, infiltration and heartbleed samples have been considered unknown attacks due to the smaller number of samples in the entire dataset.

UNR-IDD: University of Nevada - Reno Intrusion Detection Dataset (UNR-IDD) [11] is an NIDS dataset with additional special feature network port statistics collected from switch or route.

5G-NIDD: This Network Intrusion Detection dataset is generated over 5G wireless networks testbed by the University of Oulu, Finland [12]. The data is collected from the base station that is connected with normal and attacker user nodes. This dataset consists of DoS and port scan attack samples.

WUSTL-IIoT 2021: The cybersecurity attacks in Industrial IoT have been collected emulated and collected from the testbed. This dataset [13] consists of nearly 93% samples from normal traffic in the total dataset whereas the remaining 7% is occupied by Reconnaissance, command injection, DoS, and backdoor attacks. Using Argus tool, the best 41 network flow-related features are identified.

IV. PRELIMINARIES

In this section, we discussed the methodologies used in the state-of-the-art of multi-tier IDS against unknown attacks detection.

A. Unsupervised Machine Learning

1) **AutoEncoder:** The zero-day attack detection system in the state-of-the-art utilized Autoencoder as an unsupervised machine learning algorithm to learn the patterns in normal network traffic to classify them from abnormal traffic. As mentioned in [10], [32], for known samples, the latent domain in the autoencoder is closer to the normal Gaussian distribution $N(0,1)$ than in the unknown samples. The design of a Zero-day attack Intrusion Detection System used Autoencoder in the following ways;

- 1) Single Autoencoder to classify the normal data using a One-Class classifier [26]
- 2) A Single Autoencoder for benign data and its encoder hidden layers are built by CNN [27]
- 3) Dual synchronized Autoencoder for normal and attack data [3]

The encoder of the AE is connected with the classifier using the latent representation z_i of the input data x_i using an activation function A_e of the encoder

$$Q_p(x^i) = A_e(Wx^i + b) \quad (1)$$

with W and b represent the weight and bias values. The encode A_e maps x^i into a latent representation or bottleneck z^i .

2) **iForest [33]:** iForest uses a partition-based anomaly isolation method other than other methods such as density-based and distance-based to build the model for normal or anomaly samples. The tree is constructed with two child nodes, i.e., binary tree to profiling the normal instances, the deviation is identified as an anomaly. The analytical model of the iForest uses the height of the isolation tree $h(x)$ for the instance x to isolate the anomaly instances from the normal. If the path of the leaf node x from the root is more, then it is classified as normal otherwise anomaly.

$$Edge = 0; h(x) = \sum_{i=root}^x Edge + + \quad (2)$$

$$c(n) = 2H(n-1) - (2(n-1)/n) \quad (3)$$

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \quad (4)$$

$$A(x, n) = \begin{cases} Anomaly; & \text{if } s(x, n) > 0.8 \\ Normal; & \text{if } s(x, n) < 0.5 \\ NoAbnormality; & \text{if } s(x, n) \approx 0.5 \end{cases} \quad (5)$$

3) **iNNE [34], [35]:** The unknown DoS/DDoS attacks were detected using isolation using the nearest neighbor ensemble model, [15]. iNNE is a distance-based method to build the ensemble model for anomaly isolation. The axis-parallel clustering in iForest induces an anomaly masking problem that reduces the anomaly detection accuracy. This problem is resolved in iNNE by forming hyperspheres for selected subsamples with size ψ . Each instance x in the subsample S is covered by hypersphere $H(x)$ centered with x and calculated radius $r(x)$. The instance x with long radius $H(x)$ is more isolated from its neighbors. If $H(x)$ intersects other hyperspheres, then x is very close to neighborhoods. The dense hypersphere contains normal instances whereas there is more chance for the sparse hypersphere formed by anomalies. In this way, the isolation score is calculated between 0 and 1 for

the x^th instance in S . For the subsample S with n number of instances, the anomaly score A for the training dataset is calculated by the average of isolation score I of each instance;

$$A = \sum_{i=1}^n \frac{I_i}{n} \quad (6)$$

The test sample t is classified as anomaly or normal by calculating the average isolation score of t sample from n number of hyperspheres $H(x)$ generated for x instances in the subsample with ψ size;

$$A(t) = \frac{\sum_{i=1}^n I(t, H(x_i))}{n} \quad (7)$$

An anomaly score $A(t)$ of test sample t is more than the threshold value then it's classified as an anomaly;

$$t = \begin{cases} \text{Anomaly,} & \text{if } A(t) > \text{threshold} \\ \text{Normal,} & \text{otherwise} \end{cases} \quad (8)$$

4) *LOF* [36]: This mechanism learns outlier patterns rather than normal patterns in the input samples. The parameter Local Outlier Factor gives the degree of isolation for the outlier objects. This outlier factor depends on the outlier-ness of the object against its k -neighborhood. The LOF of an object is calculated via the sequence of calculations such as k -distance of an object (k -dis), reachability distance of an object (r -dis), reachability density (r -den) and outlier factor (f). The LOF is calculated by the following combined formula;

$$LOF(p) = \frac{\sum_{o \in N_M \text{ in } Pts(p)} \frac{\sum_{o' \in N_M \text{ in } Pts(p)} \text{Reach_Dist}(p, o)}{\sum_{o' \in N_M \text{ in } Pts(o)} \text{Reach_Dist}(o', o)}}{|N_M \text{ in } Pts(p)|} \quad (9)$$

$$\text{Reach_Dist}(p, o) = \max(k_{dist}(o), d(p, o)) \cdot p, o \in D \quad (10)$$

LOF finds outliers using local analysis rather than global analysis from k -nearest neighbors of an object. MinPts represents the nearest neighbors of an object.

B. Semi-Supervised Machine Learning

1) *One-Class Learner(OC-SVM)*: In the hierarchy intrusion detection system for an unknown attack [10], [28], the anomaly samples are classified from benign traffic in the first phase. The labeled benign samples train the model to learn the one-class pattern in the dataset. Gaussian kernel form the hypersphere around similar samples in the feature space instead of forming hyperplanes in default SVM. The complex, non-linear boundary builds for hyper_sphere in the OCSVM. In these works, the hyperparameters of OCSVM are optimized to induce the best AOC for anomaly detection. The one-class RBF of OCSVM is given as follows:

$$K(x_i, x_j) = \exp^{-||X_i - X_j||^2 / 2\sigma^2} \quad (11)$$

C. Supervised Learning Algorithm

1) *Machine Learning Algorithm*: One of the layers in the hybrid Intrusion Detection System included [10], [14], [30] to detect known attacks in networks that classify unknown attacks from them using the multiclass classifier

algorithms. The multi-tiered IDS [14] has been designed to detect unknown cyberattacks on the Internet of Vehicles. In the Four-tiered IDS, the first and second layers were implemented using multiclass supervised learning algorithms with hyper-parameters optimization to detect known attacks. Four tree-based learners with Tree Parzen Estimator Bayesian optimization algorithm have been stacked in the first two layers of the IDS. The Decision Tree, Random Forest, Extra Tree, and Extreme Gradient Boosting are the base learners in the ensemble model whereas high accuracy learner from this set is used as the meta classifier. The base models RF, ET and XGBoost are ensemble models of multiple decision trees. The Random Forest Supervised machine learning algorithms such as AdaBoost, Logistic Regression and Naive Bayes are adopted for the zero-day attack detection IDS for the IoT networks [30].

2) *Deep Learning*: Deep Learning model framework has been designed [8] with the help of DOC and OpenMax to classify known and unknown samples in the repeated training along with labeling via clustering technique.

Deep Open Classification [37]: Instead of, training the model in the closed environment, the learning process is done in a dynamic and open environment in the DOC, an extension of Open-class SVM. DOC should identify the new class samples that are not used in the model learning phase is called an open classification problem. DOC generates $M_{(i+1)}$ one-vs-rest models for i number of seen class labels l_i in the training dataset D and additional label for rejection class that represents unseen samples.

$$\hat{y} = \begin{cases} \text{Reject,} & \text{if } \text{Sigmoid}(d_i) < t_i, \forall i \in l_i \\ \max(\text{Sigmoid}(d_i)), \forall i \in l_i & \text{otherwise} \end{cases} \quad (12)$$

The Open Space risk of Sigmoid function is reduced by adjusting the threshold value t_i equivalent to 0.5.

OpenMax [5], [23]: The penultimate layer of the Softmax layer gives the score for an input that is not specific to any known classes. This score is called the activation vector, which is considered as the main information for OpenMax. Using the Meta-Recognition technique, OpenMax finds the distance of input from known classes. If the input is far away from known data, then it is classified as unseen sample. The Softmax layer computes the probability of detection under class label j for an input x ;

$$P(y = j|x) = \frac{e^{v_j(x)}}{\sum_{i=1}^N e^{v_i(x)}} \quad (13)$$

$$y^* = \arg\max_j P(y = j|x) \quad (14)$$

$$\hat{y} = \begin{cases} \text{Reject,} & \text{if } y^* == 0 \text{ (or)} P(y = y^*|x) < \epsilon \\ y^*, & \text{otherwise} \end{cases} \quad (15)$$

CNN: The Convolutional Neural Networks [3], [25] and LSTM are ensembled with Auto Encoder to detect unknown, known and benign classes, [4]. The ensemble model considered threshold-based design to deal the open-set problems. The CNN architecture with L1 and L2 regularizations is used in the IDS [30]. By reducing the over-fitting problem, the detection of a zero-day attack has been improvised by L1 and L2 regularization techniques. L2 performs better when

compared to L1 since L2 uses the square value of magnitude rather than absolute value in L1. The regularizers integrate the additional term, that is, penalty, with the cost function of CNN model;

$$L1 = Loss(\hat{y}, y) + \frac{\lambda}{2m} \sum_{i=1}^m ||w_i|| \quad (16)$$

$$L2 = Loss(\hat{y}, y) + \frac{\lambda}{2m} \sum_{i=1}^m ||w_i||^2 \quad (17)$$

The L2 considers all features to tune the parameters by calculating the square of magnitude that makes the less coefficient features nearer to 0 rather than 0. At the same time, the L1 method directly removes the less important features that have 0 value from the model construction process. Hence, L1 fits for outlier analysis, whereas L2 fits for complex network traffic with uncertain features to design an IDS against zero-day attacks.

Soft Ordering CNN: The traditional CNN architecture is modified in SOCNN by including the feature expansion layer before to feature extraction and classification layer to classify unknown DoS/DDoS attack detection system for IoT networks [15]. The fully connected layer is added first in the SOCNN architecture to extend the input vector x into an output matrix in $n \times m$ size where n is the existing features and m is the soft ordering feature set. The intention of SOCNN is re-ordering the features to find an optimal order with high detection rate of unknown attacks.

$$FC_{out} = g(Wx + B) \quad (18)$$

where, FC_{out} is the output matrix of fully connected layer with $n \times m$ size. $g()$ is the activation function and B is the bias value. W is the m matrix in $n \times n$ size.

LSTM [4] and BiLSTM [9]: The LSTM-based Evaluation and Target Net models of the ADQN NIDS [9] framework are used to calculate the Q value using $Q(\text{current_state}, a), \forall a \in \text{action_space}$ and updated Q value after the reward has been calculated. ADQN NIDS is designed to learn spatial, temporal and spatio-temporal features in the input samples by integrating BiLSTM, CNN and fusion of these two architectures connected via fully connected layer.

D. Reinforcement Learning

Deep Q Network combined with BiLSTM [9] selects the zero-day attack samples for annotation along with the probability distribution before classifying them using the ADQN algorithm. Using DQN [38], the dynamic active learning process is featured in IDS to retrain the model with newly annotated unknown attack samples that will improve the overall performance of the detection. The reward function of ADQN is defined as follows with two different action spaces A_k , $\forall k$ - class labels in the dataset and A_E - expert's requisition for label annotation for unknown sample;

$$r_x = \begin{cases} R_{correct}, & \text{if } a_x \notin A_E \text{ and } \hat{y}_x \equiv y_x \\ R_{wrong}, & \text{if } a_x \notin A_E \text{ and } \hat{y}_x \neq y_x \\ R_{modify}, & \text{if } a_x \in A_E \end{cases} \quad (19)$$

where, $R_{correct} = +1$, $R_{wrong} = -1$ and $R_{modify} = 0$ represent the action for correct, wrong classification and expert's request for class labeling for an input sample x . $\hat{y}_x$ and y_x means the predicted and actual class value. The LSTM-based Evaluation and Target Net models of this framework are used to calculate the Q value using $Q(\text{current_state}, a), \forall a \in \text{action_space}$ and updated Q value after the reward has been calculated.

E. Hyper-Parameter Optimization [39] (HPO)

The hyperparameters of the model are improved by two different variations of Bayesian Optimization algorithms; Tree-structured Parzen Estimator [16] and Gaussian process [40]. In the [10] research work, TPE is used as a sampling algorithm. The k -value in the k -means unsupervised clustering algorithm is tuned using the GP-BO method in the work [14]. The Gaussian distribution in GP-BO method predicts the class label y for an input x in the hyper-parameter space D with mean μ and covariance σ .

$$p(y|x, D) = N(y|\hat{\mu}, \hat{\sigma}^2) \quad (20)$$

The prediction of next hyper-parameter value in TPE is calculated using two density functions $l(x)$ and $g(x)$ with the predefined threshold y^* . These two functions are defined for good region and poor region which is adjusted by $l(x)/g(x)$ to obtain optimal values for hyper-parameters.

$$p(x|y, D) = \begin{cases} l(x), & \text{if } y < y^* \\ g(x), & \text{if } y > y^* \end{cases} \quad (21)$$

V. IMPLEMENTATION MECHANISM

In this section, we discuss the proposed enhanced IDS to detect unknown attacks and compare it with the existing multi-tier hierarchical IDS [10]. We have considered Verkerken et al. [10] IDS for an empirical study over different networks based IDS datasets. This IDS is a three-tier, multi-stage framework to detect benign, known, and unknown attack samples using anomaly-based and signature-based approaches. Totally three stages namely anomaly detection, known attack detection and unknown attack detection are used to build this hierarchical IDS. To improve the unknown attacks classification accuracy, anomaly scores generated in stage1 and stage2 for the test samples are compared with the corresponding phase revised thresholds τ_B , τ_M and τ_V of the unsupervised and supervised classification algorithms rather than their default threshold values. The anomaly detector model's threshold τ_B is calculated from the maximum f-score in the F1 to F9 scores of the classification algorithm. In the second stage, F1 score of the multi-class classification algorithm is set as threshold τ_M . This work compared and identified One Class Support Vector System algorithm performs better than Auto Encoder for an unsupervised algorithm to classify benign samples in the anomaly detection stage. Whereas, Random Forest performs better than Neural Networks for a supervised algorithm to classify the known attacks in the multiclass classification stage. The performance of the stage1 and stage2 classification models is improved by the optimization of the hyper-parameters using the TPE method. The CIC_IDS_2017

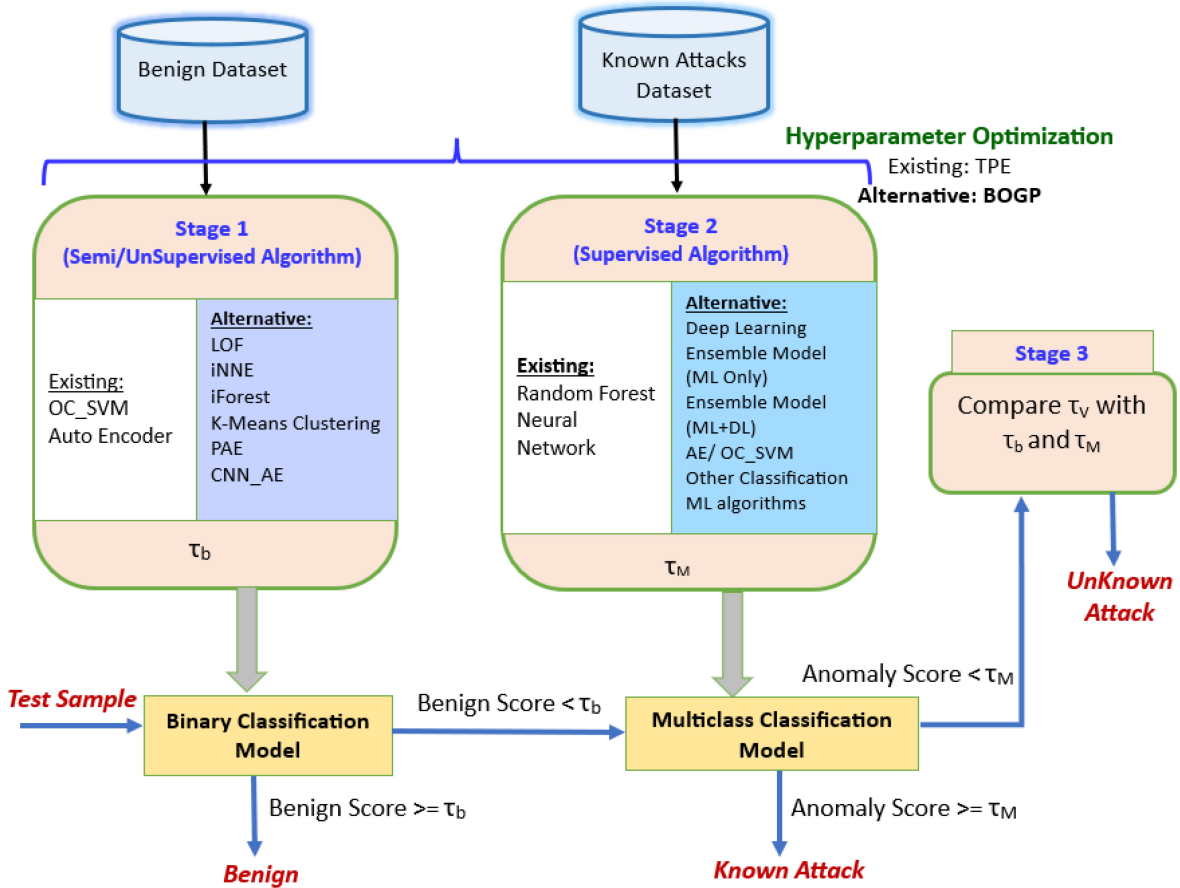


Fig. 2. The Proposed Enhanced Multi-tier Hierarchical IDS Architecture with various Implementation Techniques.

dataset has been considered to evaluate the IDS performance. The major drawbacks we have identified in the existing work are (i) the efficiency of this system has not compared with other multi-tier zero-day attack detection systems. This system is compared only with simple open-set IDS which tuned the RF baseline threshold (ii) other intrusion detection benchmark datasets have not been considered to analyze the IDS performance under different characteristics of various networks and (iii) it can't achieve the high classification accuracy for zero-day attacks.

We initiated our work by considering this setback in Verkerken et al. [10] work and the notable technologies in other zero-day attack detection systems which are discussed in Sections II and III. In our work, we adapted the basic structure of Verkerken et al. [10] IDS with an adjustable threshold since the framework with clustering and classification is the best one for unknown attack detection in open-set problems. The novelty in our work is the multi-stage hierarchical IDS is designed against zero-day attacks with different algorithms mentioned in the state-of-the-work. The proposed framework shown in Fig. 2 is designed with three stages such as anomaly detection, known attack detection, and unknown attack confirmation which is similar to existing hierarchical IDS. The enhancement is introduced via two ways; (i) find the best combination from the permutation of clustering and classification algorithms to design the open-set IDS to provide a general solution and (ii) BOGP optimized

models. The anomaly detector model is trained using the alternative methods LOF, iNNE, iForest, k-means clustering, PAE, and CNN with AE to classify the samples as benign or not. In stage2, the multi-class classification model is build using various algorithms and ensemble techniques to classify known attacks. The sample which is with less anomaly score scored in stage2 than the stage2 threshold value τ_M then it forwards to stage3 otherwise this is classified as known attack by the stage2. In the final stage, the anomaly score of unclassified samples is compared with stage1 and stage3 threshold values. If the anomaly score generated in stage2 for unclassified samples is less than stage3 threshold τ_V then it is classified as an unknown attack. The BOGP is applied to tune the hyper-parameters of the models in both stages.

The existing and enhanced IDS (represented as “Alternative” in Fig. 2) performance is analyzed using various intrusion detection benchmark datasets such as CIC_IDS_2017, 5G, UNR and WUSTL. To introduce an unknown attack in the dataset for implementation, a known attack class with less occurrence is considered for that purpose which is mentioned in Table II.

VI. RESULTS AND DISCUSSION

A. Implementation Environment and Performance Metrics

We used high end GPU platform, NVIDIA H100 80GB PCIe 5.0 x 16 for the implementation of hybrid, multi-tier

TABLE II
DETAILS OF UNKNOWN ATTACK SAMPLES IN THE IMPLEMENTATION OF THE PROPOSED ENHANCED MULTI-TIER IDS

Dataset	Benign	#Known	#Unknown	Known Labels	Unknown Labels	Total Labels(Attack + Benign)
5G	475279	735593 (Sampled 3000)	1155 (0.000952949%)	7 (UDPFlood, HTTPFlood, SlowrateDoS, SYNScan, TCPConnectScan, UDPScan, SYN-Flood)	1 (ICMPFlood)	8+1
WUSTL	1107448	86545 (Sampled 8240)	471(0.00042074%)	2 (DoS, Reconn)	2 (CommInj, Backdoor)	4+1
UNR	3773	32615 (Sampled 5615)	1022 (0.027318899%)	4 (PortScan, TCP-SYN, Blackhole, Diversion)	1 (Overflow)	5+1
CIC_IDS_2017	2095057	425694 (Sampled 1948)	47 (0.000018645%)	5 (Botnet, BruteForce, DDoS, WebAttack, PortScan)	2 (Heartbleed, In-filtration)	7+1

IDS with optimized parameters. It is the most powerful GPU chip on the market and is designed specifically for artificial intelligence (AI) applications. The configuration details of H100 is with the processors in 2 x 24-core (48 cores), memory in 1TB, OS storage in 1 x 1.9TB NVMe PCIe, file storage in 3 x 14TB SATA 6Gb/s 7.2K and NVIDIA NVLink Bridge. The four datasets CIC_IDS_2017, WUSTL, 5G and UNR IDD intrusion detection benchmark datasets of different networks are stored in the server and executed using the python code with appropriate packages such as PyTorch and TensorFlow etc. The metrics such as accuracy, precision, recall, balanced accuracy, and F-score have been measured to analyze the performances of the hierarchical intrusion detection system. Out of that, the formula for balanced accuracy and F-score are given below:

Balanced accuracy: This is one of the vital metrics for highly imbalanced datasets by considering the accuracy of both positive and negative cases. The datasets used in our experiments are highly imbalanced. We have not adapted any techniques to balance the actual dataset since they reduce the quality of the dataset. Hence, balanced accuracy is the perfect measure to analyze the system under this case.

$$\text{Balanced Accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2} \quad (22)$$

F score: It combines Precision and Recall to analyze the performance of the algorithm.

$$F_{\beta} = \left(1 + \beta^2\right) \frac{\text{Precision} \cdot \text{Recall}}{(\beta^2 \cdot \text{Precision}) + \text{Recall}} \quad (23)$$

where, β varies from 1 to 9 for calculating F1 score to F9 score.

B. Performance Evaluation and Results Analysis

The existing multi-tier hierarchical IDS [10] is implemented and evaluated its' performance using various benchmark intrusion datasets such as UNR, 5G and WUSTL. In three stages of the IDS architecture, the performance of the machine learning algorithm is observed for an evaluation of binary and multi-classification models to detect normal, known and unknown attacks. Using various benchmark datasets, the pros and cons of the existing hybrid IDS design are well analyzed. The metrics such as training time, test time, accuracy, balanced accuracy, precision, recall and F scores are identified as appropriate to evaluate the performance of the design of an

TABLE III
GENERATED THRESHOLDS BY THE EXISTING IDS [10]

Dataset	τ_b (stage1)	τ_m (stage2)	τ_u (stage3)
5G	0.207	0.950	-0.003
WUSTL	0.170	0.950	-0.113
UNR	0.072	0.700	-0.256
CIC_IDS_2017	0.001	0.914	0.995

IDS to detect unknown attacks along with benign and known attacks.

The implementation in our work has been carried out for three major observations:

- 1) Existing multi-tier IDS vs other networking benchmark IDS datasets: These results evaluate the efficiency of the existing multi-tier IDS using different IDS datasets
- 2) The Proposed Enhanced multi-tier IDS: The techniques used in the state-of-the-art of zero-day attack detection system are applied in anomaly detection and multi-class attack detection stages in the Existing multi-tier IDS
- 3) BOGP vs TPE Optimization techniques: The Proposed Enhanced multi-tier IDS is evaluated with a different optimization technique BOGP than the proposed one TPE to tune the hyper-parameters of the anomaly detection and classification models

We present the implementation results in the following sections to discuss the efficiency of the proposed enhanced IDS along with TPE and BOGP optimization techniques and the limitations of the existing work on various datasets.

C. Existing Multi-Tier IDS vs Various Benchmark IDS Datasets

In this section, the limitations of the existing multi-tier IDS [10] are discussed in stage-wise detail with the help of various IDS datasets. The limitations outlined here emphasize the significance of the general open-set approach-based IDS framework against unknown attacks regardless of the features specific to the datasets. Primarily, we focused on the execution time and classification performances to analyze the performance of the IDS on various datasets and presented these results in Table III to Table V, and Fig. 3–Fig. 6.

1) *Execution Time of the IDS*: The training time of stage1 and stage2 and the testing time of all three stages of the existing system are shown in Table V. The stage1 training time of WUSTL is recorded as high, i.e., 1min 39s $\pm$ 924 ms and CIC_IDS_2017 is observed as low 0.771 s $\pm$ 21 ms

TABLE IV
PERFORMANCE METRICS OF EXISTING MULTI-TIER HIERARCHICAL IDS

DataSet	Full System											
	Precision			Accuracy (%)			Recall			F1 Score		
	Unknown Attack	Known Attack	Benign	Unknown Attack	Known Attack	Benign	Unknown Attack	Known Attack	Benign	Unknown Attack	Known Attack	Benign
CIC_IDS_2017	0.069	0.924	0.99	87.2	93.44	98.6	0.87	0.93	0.986	0.13	0.93	0.99
UNR	0.20	1.0	1.0	99.8	76.3	99.7	1.0	0.76	1.0	0.33	0.87	1.0
WUSTL	0.47	1.0	1.0	82.8	97.9	99.6	0.83	0.98	1.0	0.60	0.99	1.0
5G	0.72	0.97	0.99	99.9	68.8	99.7	1.0	0.69	1.0	0.83	0.80	0.99

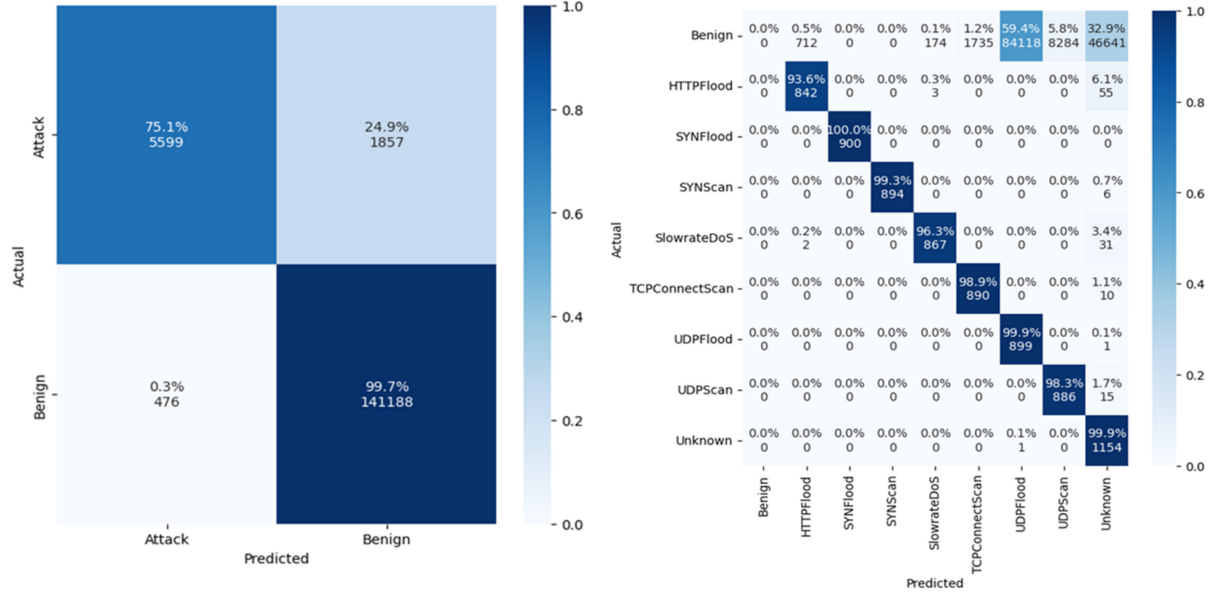


Fig. 3. Confusion Matrix of Stage1 and Stage2 of Existing IDS for 5G Dataset.

TABLE V
TIME COMPLEXITY OF PROPOSED IDS VS EXISTING IDS [10]

Dataset	Stage1 Training Time	Stage1 Test Time	Stage2 Training Time	Stage2 Test Time	Combined (stage1 & stage2) Test Time
Proposed IDS					
WUSTL	1min 39s ± 924 ms	1.45 s ± 40.71 ms	4.43 s ± 46 ms	646.9 ms ± 11.48 ms	1.707 s ± 56.52 ms
UNR	9.59 s ± 662 ms	122.6 ms ± 20.75 ms	50.6 s ± 7.44 s	26.84 ms ± 992.2 μs	54.17 ms ± 4.985 ms
5G	40 s ± 380 ms	4.708 s ± 5.151 ms	19.8 s ± 823 ms	1.813 s ± 8.387 ms	5.65 s ± 43.49 ms
CIC_IDS_2017	0.67 s ± 7.8 ms	4.7 s ± 158 ms	2.8 s ± 1.2 ms	0.3 s ± 2 ms	4.2 s ± 13 ms
Existing Multi-tier IDS [10]					
CIC_IDS_2017	0.771 s ± 21 ms	8.117 s ± 277 ms	1.249 s ± 034 ms	0.610 s ± 5 ms	8.043 s ± 65 ms

out of 4 considered datasets. The WUSTL dataset contains 11,07,448 benign samples to train the OC_SVM model which is higher than UNR and 5G datasets that induce the high training time for WUSTL. Interestingly, the test time is high for CIC_IDS_2017 with $8.117 \text{ s} \pm 277 \text{ ms}$ and UNR secures low with $122.6 \text{ ms} \pm 20.75 \text{ ms}$ due to how complex the model converges to find the solution. The known attacks detected in $1.813 \text{ s} \pm 8.387 \text{ ms}$ for 5G dataset that is the high value observed in the 4 datasets. Similarly, the unknown classification takes more time in 5G and CIC_IDS_2017

datasets such as $5.65 \text{ s} \pm 43.49 \text{ ms}$ and $8.043 \text{ s} \pm 65 \text{ ms}$ respectively. It shows how complex these two datasets are to derive the unknown attacks.

2) *Anomaly Detection Phase*: The unsupervised and semisupervised algorithms correspondingly AE and OCSVM are used in the anomaly detection model of the existing system and identified OCSVM outperforms AE. The classification model is optimized by TPE technique. Hence, we implemented the first stage of the existing system using OCSVM and AE and then observed the high value generated by these two methods. The confusion matrix generated in stage1 for the datasets 5G, WUSTL and UNR are presented in Fig. 3, Fig. 4 and Fig. 5 correspondingly. From Fig. 3, the accuracy of anomaly detection in stage1 for the 5G dataset is observed very less, i.e., 75% than for other two datasets WUSTL and UNR which are shown in Fig. 4 and Fig. 5. For that two datasets, anomaly labels are classified with more than 92%.

Table VI shows the other performance metrics such as recall, precision and F1-score in stage1 for all four datasets. Correspondingly, the Recall value of the attack class shown in Table VI reached 79% for the 5G dataset which is lower than in other datasets whereas the precision is comparatively lesser than others. The anomaly samples are wrongly classified as benign which reduces the recall of attack and precision of benign samples for UNR and WUSTL. Another interesting observation noticed in the stage1 results is the dataset CIC_IDS_2017 secured less accuracy 68.7%

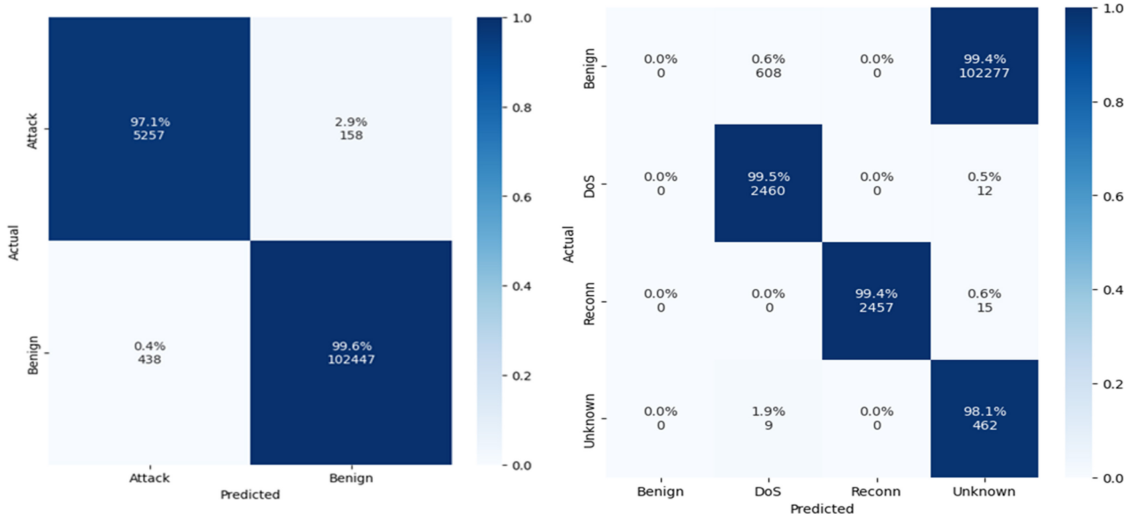


Fig. 4. Confusion Matrix of Stage1 and Stage2 of Existing IDS for WUSTL Dataset.

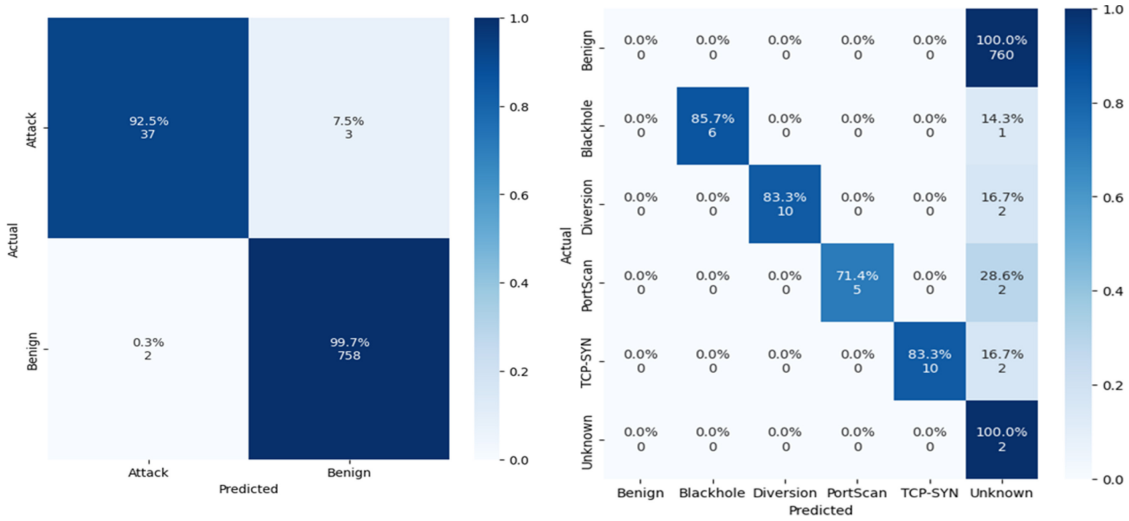


Fig. 5. Confusion Matrix of Stage1 and Stage2 of Existing IDS for UNR Dataset.

for benign sample prediction. Correspondingly, the recall of benign samples and precision of attack samples are observed as less than the other datasets. The reason behind this is more number of benign samples in the CIC_IDS_2017 dataset are wrongly classified as attacks. That reduces the recall of benign and the precision of attack classes. In general, OCSVM in stage1 performs better for UNR and WUSTL datasets than 5G and CIC_IDS_2017 datasets for attack and benign sample detection.

3) *Multiclass Known Attack Detection Phase*: The major design intention of stage2 is to classify known attacks with high accuracy. The multiclassification algorithm, Random Forest used to train the Stage2 classifier. The threshold τ_m in this stage is calculated from the high score registered in the range F1 to F9 scores. Refer Table III for the threshold obtained in stage2 for different datasets. The confusion matrix of 5G, WUSTL and UNR are shown in Fig. 3, Fig. 4 and Fig. 5 respectively. The low accuracy 80.93% registered for known attacks in UNR dataset due to its small size dataset than other datasets. Simultaneously, UNR registered less recall 81% for

known attacks since more known attack samples are wrongly classified into other classes like benign. More negative cases are wrongly classified as known attacks in 5G dataset which causes low precision 0.55 for this dataset.

4) *Unknown Attack Detection Phase/Extension Stage*: This stage finalizes the unknown attacks. The known attack is classified from benign by stage1 and then stage2 confirms it. Table VI shows the overall performance of the existing multi-tier IDS. The classification accuracy for unknown attacks for CIC_IDS_2017 and WUSTL datasets are registered less values 87% and 83%. More benign and known attack samples are wrongly classified as unknown attacks in CIC_IDS_2017, UNR and WUSTL datasets which reduces the precision to less than 0.5. At the same time, more unknown attack samples are wrongly classified which reduces the recall value near to 0.8. Fig. 6 shows the confusion matrix of the complete system of UNR, WUSTL and 5G datasets. Table III shows the threshold values of each stage. The threshold value τ_b is calculated from F-scores obtained from stage1 prediction algorithm. τ_b is observed as low 0.00047 in CIC_IDS_2017

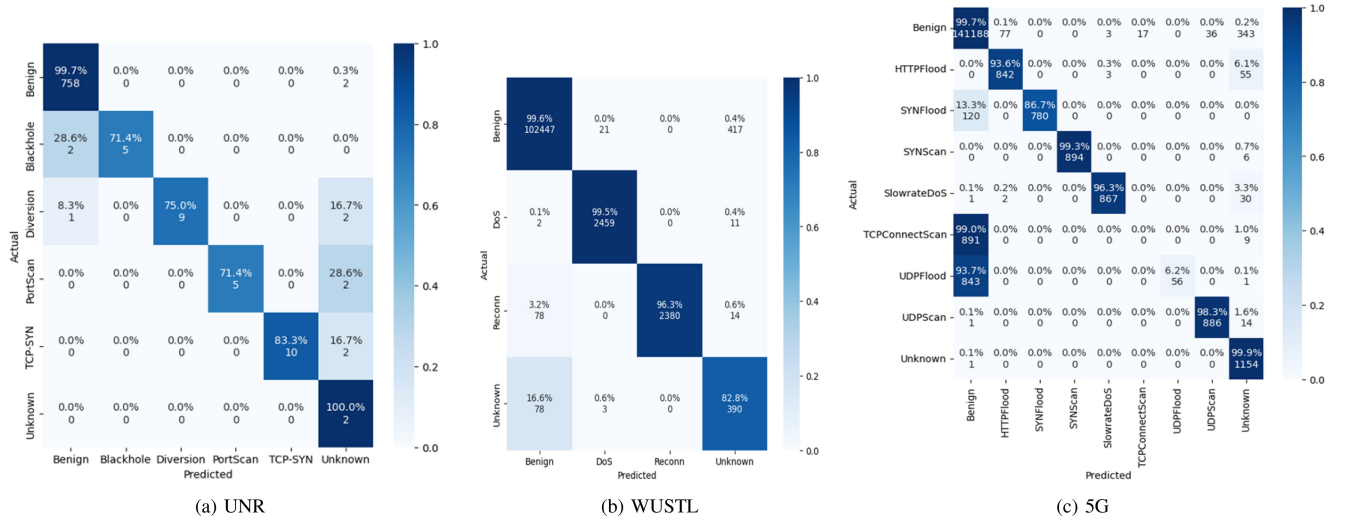


Fig. 6. Confusion Matrix of the Complete System in the Existing IDS for (a) UNR (b) WUSTL and (c) 5G.

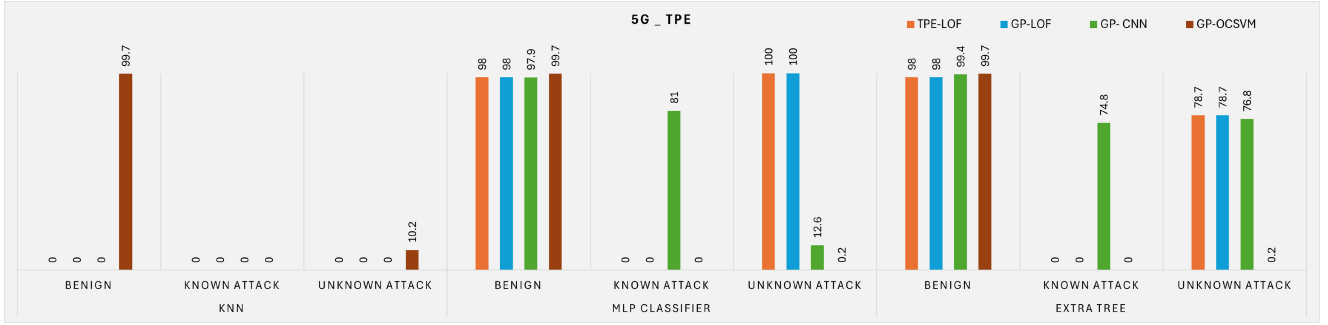


Fig. 7. The Proposed Enhanced Multi-tier IDS for 5G with TPE.

and high 0.2068 in 5G dataset to classify the attacks from benign.

D. Performance Analysis of the Proposed Enhanced Multi-Tier Hierarchical IDS

This section applies different techniques from state-of-the-art for anomaly detection and multi-class classification under deep learning and machine learning. For stage1 of multi-stage IDS, anomaly detection is implemented using unsupervised algorithms such as LOF, iForest, iNNE, CNN with AE, k-means, one-class SVM under two different model optimization techniques TPE and BOGP. The implementation results are shown in Table VI. The top 5 algorithms are chosen from Table VI to implement stage1 in the multi-stage IDS. For UNR and WUSTL datasets, LOF_TPE, LOF_GP, iForest_GP, and OC-SVM_GP are best algorithms for anomaly detection to classify benign and non-benign samples. The algorithm iForest_TPE is an additional one for WUSTL dataset whereas iNNE_GP is in the top list of UNR. The top 5 performers of 5G dataset are LOF_TPE, LOF_GP, CNN_GP, iNNE_GP and OC-SVM_GP and for CIC_IDS_2017 dataset, CNN_TPE, CNN_GP, iNNE_GP, and OC-SVM_GP.

In the second stage, known attacks are classified using the following supervised algorithms from deep learning CNN, Soft CNN, LSTM, GRU and all types of multi-class classification

algorithms from machine learning. The ensemble machine learning models like AdaBoost, XGBoost etc and customized ensemble model from deep learning and machine learning under bagging are also used to implement the stage2 of multi-stage IDS. The results of complete IDS for 5G, UNR, WUSTL and CIC_IDS_2017 datasets are shown in Fig. 7 to Fig. 12 along with different optimization algorithms TPE and BOGP.

For UNR dataset that shown in Fig. 9 and Fig. 10, under BOGP optimization algorithm nearly 75% overall accuracy scored by the J48G machine learning algorithm in stage2 for known attacks classification and LOF_TPE as well as iForest_GP algorithms in stage1 for benign samples detection. The unknown attacks are identified in 99.94% but benign and known attacks samples are classified with only 50%. Hence, we considered the next low-level top performers GRU, MLP and XGB with LOF_TPE or LOF_GP secured 82% known and benign samples classification accuracy. In this set of algorithms, XGB and MLP provides nearly 30% detection accuracy for unknown attacks with LOF_TPE and LOF_GP algorithms in stage1 correspondingly. For the UNR dataset shown in Fig. 9 with TPE model optimization algorithm, the stage2 algorithms such as MLP, CNN and XGB provide high benign and known attacks accuracy in the final detection result like 99.97% and 68% correspondingly. The algorithms such as LOF_TPE, LOF_GP, iForest_GP are best combination in stage1 along with a set of selected stage2 algorithms. Finally,

TABLE VI
STAGE1 PERFORMANCES IN THE EXISTING IDS AND THE PROPOSED ENHANCED IDS

DataSet	Stage 1							
	Precision		Recall		F1-Score		Accuracy (%)	
	Benign	Attack	Benign	Attack	Benign	Attack	Benign	Attack
Existing Multi-tier IDS [10]								
CIC_IDS_2017	0.996	0.14	0.69	0.96	0.82	0.24	68.7	96.0
UNR	0.996	0.95	0.997	0.93	0.996	0.94	99.7	92.5
WUSTL	0.998	0.92	0.996	0.97	0.997	0.94	99.6	97.1
5G	0.987	0.92	0.997	0.75	0.992	0.83	99.7	75.1
The Proposed Enhanced IDS								
TPE Optimization + LOF								
CIC_IDS_2017	0.99	0.16	0.76	0.87	0.86	0.27	75.4	86.8
UNR	0.99	0.99	0.99	0.98	0.99	0.99	99.7	97.8
WUSTL	0.99	0.82	0.99	0.99	0.99	0.90	98.9	99.9
5G	0.99	0.89	0.97	0.96	0.98	0.92	97.5	96.0
TPE Optimization + iForest								
CIC_IDS_2017	0.99	0.11	0.59	0.96	0.75	0.20	59.2	95.6
UNR	0.99	0.80	0.89	0.99	0.94	0.89	89.3	99.4
WUSTL	0.99	0.65	0.97	0.99	0.99	0.79	97.2	100.0
5G	0.98	0.47	0.79	0.93	0.87	0.63	78.5	93.3
TPE Optimization + iNNE								
CIC_IDS_2017	0.7	0.34	0.4	0.2	0.38	0.24	56.3	96.0
UNR	0.63	0	0.74	0	0.68	0	74.3	0.0
WUSTL	0.99	0.28	0.86	0.99	0.93	0.43	86.3	99.9
5G	0.96	0.69	0.92	0.83	0.94	0.75	92.3	83.2
TPE Optimization + CNN_AE								
CIC_IDS_2017	0.97	0.37	0.96	0.43	0.97	0.40	96.2	42.7
UNR	1	0.75	0.86	1	0.93	0.86	86.1	100.0
WUSTL	1	0.55	0.96	1	0.98	0.71	95.7	100.0
5G	0.97	0.80	0.96	0.85	0.96	0.82	95.7	84.9
TPE Optimization + kMeans								
CIC_IDS_2017	0.99	0.11	0.63	0.82	0.77	0.19	63.1	82.0
UNR	0.78	0.41	0.66	0.57	0.71	0.48	65.7	56.6
WUSTL	0.96	0.06	0.61	0.49	0.75	0.11	61.2	48.6
5G	0.88	0.39	0.87	0.41	0.87	0.40	86.9	41.4
BO_GP Optimization + LOF								
CIC_IDS_2017	0.99	0.15	0.75	0.87	0.85	0.26	74.6	86.5
UNR	0.99	0.93	0.97	0.99	0.98	0.96	96.8	99.7
WUSTL	1	0.83	0.99	1	0.99	0.91	98.9	100.0
5G	0.99	0.89	0.97	0.96	0.98	0.92	97.5	95.6
BO_GP Optimization + iForest								
CIC_IDS_2017	0.99	0.12	0.69	0.82	0.81	0.21	69.2	82.3
UNR	0.98	0.82	0.91	0.97	0.94	0.88	90.7	96.6
WUSTL	0.99	0.65	0.97	0.93	0.99	0.77	97.4	93.2
5G	0.98	0.43	0.74	0.94	0.84	0.59	73.8	94.2
BO_GP Optimization + iNNE								
CIC_IDS_2017	0.34	0.1	0.3	0.1	0.31	0.09	18	96.0
UNR	0.99	0.91	0.96	0.97	0.97	0.94	95.9	97.2
WUSTL	1	0.12	0.62	1	0.77	0.22	62.4	100.0
5G	0.96	0.99	0.99	0.82	0.98	0.89	99.8	81.7
BO_GP Optimization + CNN_AE								
CIC_IDS_2017	0.97	0.30	0.94	0.47	0.96	0.37	94.4	46.6
UNR	1	0.77	0.87	1	0.93	0.87	87.4	100.0
WUSTL	1	0.53	0.95	1	0.98	0.69	95.3	100.0
5G	0.97	0.82	0.96	0.85	0.96	0.83	96.1	84.8
BO_GP Optimization + kMeans								
CIC_IDS_2017	0.98	0.07	0.46	0.83	0.63	0.14	46.2	83.4
UNR	0.76	0.40	0.67	0.51	0.72	0.45	67.2	51.4
WUSTL	0.96	0.07	0.68	0.44	0.7	0.4	68.4	44.2
5G	0.89	0.35	0.80	0.53	0.84	0.43	80.1	53.3
BO_GP Optimization + OCSVM								
CIC_IDS_2017	0.98	0.34	0.94	0.57	0.96	0.43	94.2	57.2.0
UNR	0.93	1	1	0.84	0.97	0.91	100	83.7
WUSTL	0.99	0.70	0.98	0.84	0.99	0.76	98.1	84.4
5G	0.94	0.98	0.99	0.68	0.97	0.80	99.7	68.1

the XGB and MLP are identified as a best stage2 algorithms for entire IDS with iForest_GP in stage1. They secured 100%:58%:38% and 100%:68%:32% for benign, known and unknown attacks in the complete multi-stage IDS.

The same kind of analysis we have applied for other datasets CIC_IDS_2017, 5G and WUSTL to find the best combination of algorithms in stage1 and stage2 along with model optimization technique. For the WUSTL dataset with TPE

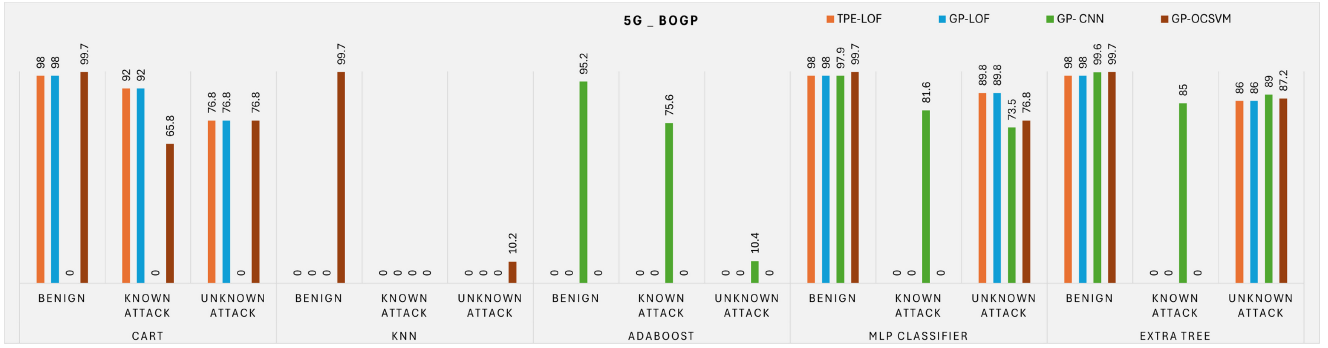


Fig. 8. The Proposed Enhanced Multi-tier IDS for 5G with BOGP.

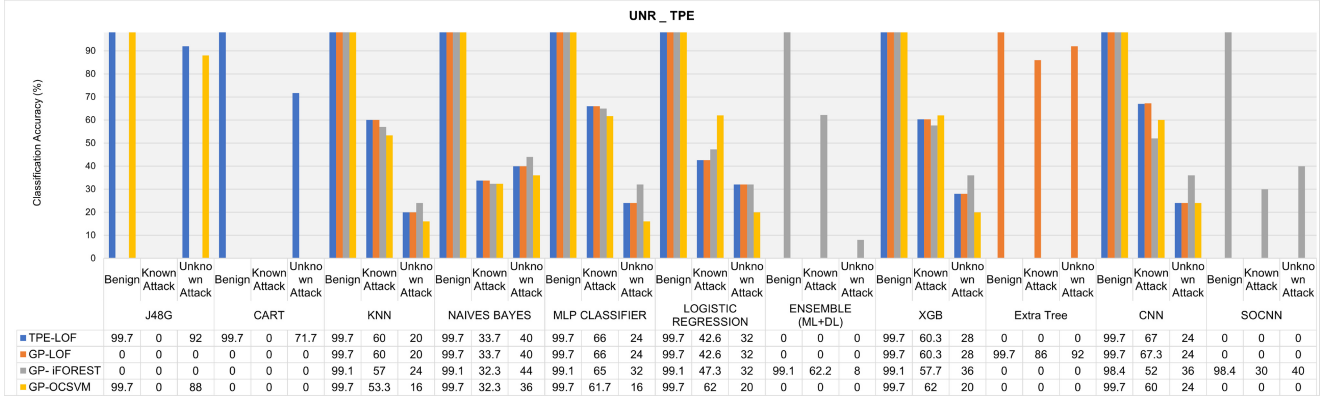


Fig. 9. The Proposed Enhanced Multi-tier IDS for UNR with TPE.

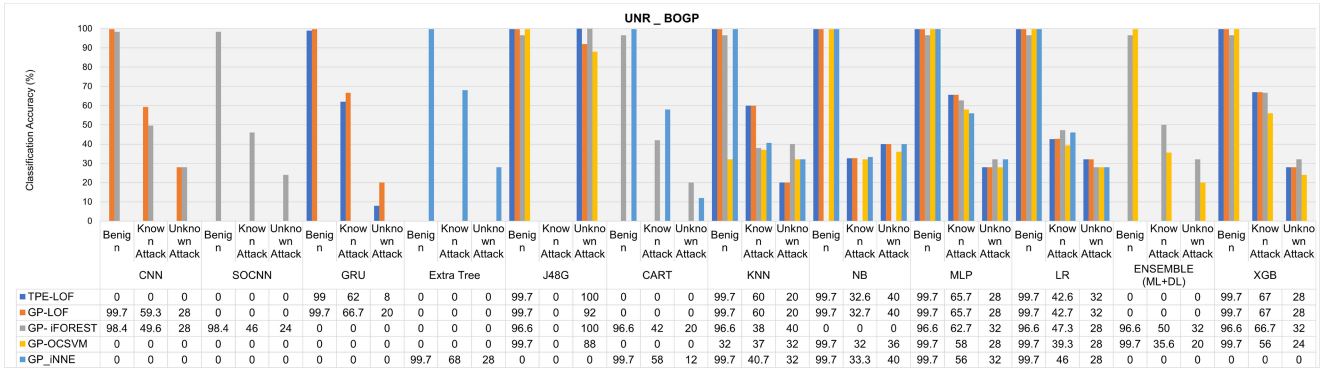


Fig. 10. The Proposed Enhanced Multi-tier IDS for UNR with BOGP.

model optimization technique shown in Fig. 11, MLP and Extra Tree are identified as best algorithm at stage2 and iForest_GP in stage1. The algorithms MLP and ET secured 98%:65%:32% and 98%:58%:38% detection accuracy in the final stage of the multi-stage IDS for benign, known attacks and unknown attack samples. For BOGP model optimization algorithm for WUSTL dataset shown in Fig. 12, the MLP and Extra Tree stage2 algorithms secure high detection accuracy in the final stage with LOF_TPE and LOF_GP algorithms in stage1. These algorithms provide 99.97%:68%:29% detection accuracy in the final stage for benign, known attacks and unknown attacks.

For 5G dataset, Extra Tree algorithm in stage2 and CNN_GP in stage1 provides best detection accuracy for benign, known attacks and unknown attacks with 99.4%, 74.8% and 76.8% respectively. This result is shown in

Fig. 7 under TPE model optimization technique. For BOGP optimization technique in the model for 5G dataset shown in Fig. 8, MLP classifier in stage2 secures high detection accuracy 97.9%:81.8%:73.5% for benign, known attacks and unknown attacks. The best combination algorithm in stage1 is CNN_GP for MLP (Multi Layer Perceptron) in stage2 to obtain this best results. For CIC_IDS_2017 dataset, Verkerken et al. [10] system provides high accuracy than other alternative algorithms in stage1 and stage2.

E. Existing Multi-Tier IDS vs the Proposed Enhanced Multi-Tier IDS

The existing multi-tier hierarchical IDS proposed by Verkerken et al. [10] performance over the various benchmark

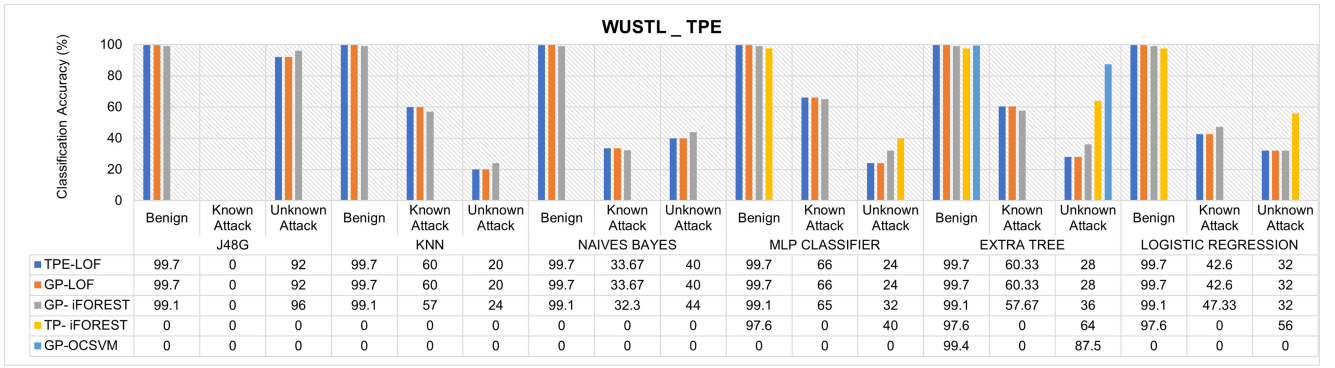


Fig. 11. The Proposed Enhanced Multi-tier IDS for WUSTL with TPE.

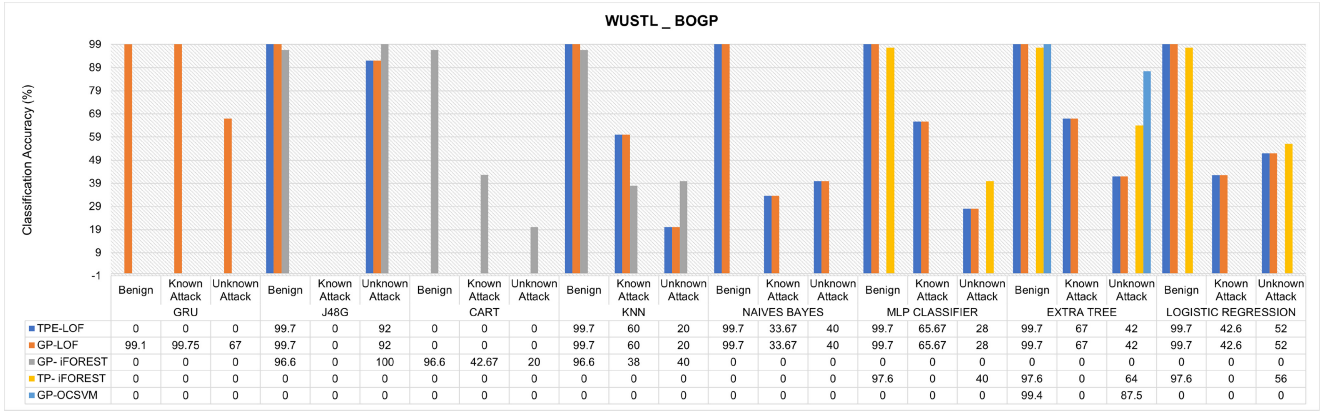


Fig. 12. The Proposed Enhanced Multi-tier IDS for WUSTL with BOGP.

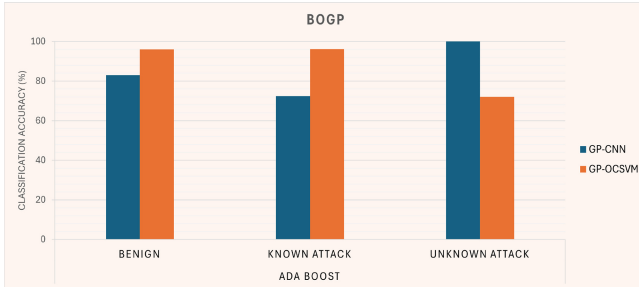


Fig. 13. The Proposed Enhanced Multi-tier IDS for CIC_IDS_2017 with BOGP.

network IDS datasets is compared with our enhanced multi-tier IDS and presented in Fig. 14. The best combination of stage1 and stage2 algorithms is identified in all four datasets under TPE and BOGP optimization algorithms for this comparison. For the 5G dataset, CNN with GP in stage1, Extra Tree with BOGP in stage2 is the best combination for the proposed enhanced IDS. For the WUSTL dataset, LOF with GP in stage1, the deep learning technique GRU with BOGP in stage2 is the best combination for enhanced IDS. For the UNR dataset, LOF with GP in stage1, Extra Tree with TPE in stage2 is the best combination for the proposed enhanced IDS. For the CIC_IDS_2017 dataset, CNN or OCSVM with GP in stage1, AdaBoost with BOGP in stage2 is the best combination for enhanced IDS. The benign samples are generally well classified in all the datasets by the existing and our enhanced IDS. Known attack samples in the enhanced IDS are classified

better than existing IDS. However, the unknown attack samples classification rate is impressive in the enhanced IDS but that are not upto the level of existing IDS. In general, the overall classification rate is high in the proposed enhanced multi-tier IDS with less computation complexity than the existing multi-tier IDS.

F. Discussions and Implications

The following points are observed from the meticulous analysis in the results generated from our implementation:

- 1) Overall, the machine learning algorithms such as MLP, Extra Tree and XGB are the best performers in the stage2 of the proposed enhanced multi-tier, hierarchical IDS
- 2) For stage1, iForest_GP, LOF_GP, LOF_TPE and CNN_GP are identified as the best algorithms to achieve high results in the final stage of our enhanced multi-tier, hierarchical IDS
- 3) BOGP model optimization technique performs well to detect benign, known attacks and unknown attacks for all 4 datasets than the TPE
- 4) From the overall observation, MLP and iForest perform better in stage2 and stage1 of our enhanced multi-tier IDS for multi-class known attacks classification and benign samples detection
- 5) For 5G and UNR datasets, the benign, known and unknown attack samples are detected in high

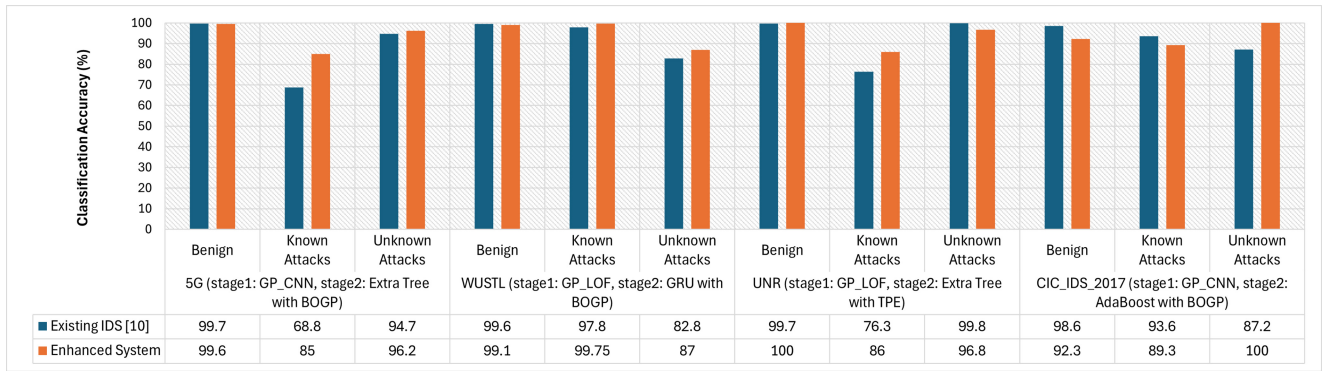


Fig. 14. The Proposed Enhanced Multitier IDS vs Existing Multitier IDS.

percentage which is more than 85% than other datasets CIC_IDS_2017, UNR and WUSTL

- 6) For stage1, the Gaussian-based optimization algorithm produces better results than TPE
- 7) For stage1, the unsupervised and semi-supervised algorithms such as LOF, OCSVM, and the deep learning technique CNN give the best results for anomaly detection
- 8) For stage2, GRU, Ada Boosta and Extra Tree perform better for multiclass classification in known attacks

VII. CONCLUSION

The traditional, single-level intrusion detection system does not perform well against zero-day attacks due to its closed-set recognition nature. Hence, the hierarchical, and hybrid nature intrusion detection systems is implemented and well-analyzed in this empirical study with different alternative techniques from the state-of-art works. The alternative techniques for stage1's unsupervised or semi-supervised learning and stage2's supervised learning are identified and observed performance metrics of the proposed IDS. We adopted different hyper-parameter optimization algorithms such as Tree-structured Parzen estimator and Bayesian optimisation with Gaussian processes for model fine tuning. These techniques improve the detection accuracy of unknown attacks especially along with threshold optimization at baseline classifier. In this empirical study, the efficiency of the proposed enhanced multitier IDS is evaluated using various IDS datasets such as CIC_IDS_2017, UNR, WUSTL and 5G. From this detailed study, we identified the best algorithms for each layer of the multi-tier IDS with suitable model optimization algorithms. The advantages of our proposed work are the design of the best unknown attack IDS over more datasets, better detection accuracy, and less computation complexity in training and inference time. The combination of suitable hybrid algorithms at multi-levels, an adjustable threshold in the classifier, hyper-parameter optimized models, zero human/expert intervention, one-shot learning, and the fusion of DL and ML techniques makes the proposed IDS to perform well on unknown-attacks. Three major drawbacks are in line for future work: (i) improve the overall detection accuracy better than the achieved one with advanced feature learning method and (ii) confront the real-time traffic in open-set recognition problem with less positive rate and inference time.

APPENDIX

NOMENCLATURE

IDSs	Intrusion Detection Systems
HIDS	Hierarchical IDS
IoT	Internet of Things
IIoT	Industrial-IoT
OSR	Open Set Recognition
TPE	Tree-structured Parzen Estimator
BOGP	Gaussian Process-based Bayesian optimisation
IoV	Internet of Vehicles
MTH	multilevel hybrid
IG	Information Gain
FCBF	Fast Correlation Based Feature Selection
KPCA	Kernel Principal Component Analysis
BO-TPE	Bayesian Optimization-Tree of Parzen Estimator
PPPL	Proportional Progressive Pseudo-Labeling
FL	Federated Learning
AE	Autoencoder
DL	Deep Learning
ML	Machine Learning
FGSM	fast gradient sign method
CW	Carlini Wagner
DoS/DDoS	Denial of Service/ Distributed DoS
DVAE	VAE-based dual-domain defense
DQN	deep Q-network
LOF	Local Outlier Factor
iNNE	isolated-based Nearest Neighbor Ensemble
SOCNN	Soft Ordering CNN
CIC	Canadian Institute of Cybersecurity
UNR-IDD	University of Nevada - Reno Intrusion Detection Dataset
OC-SVM	One-Class Support Vector Machine
HPO	Hyper-parameter Optimization
GP	Gaussian Process
MLP	Multi Layer Perceptron

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REFERENCES

- [1] U. Sabeel, S. S. Heydari, K. El-Khatib, and K. Elgazzar, "Unknown, atypical and polymorphic network intrusion detection: A systematic survey," *IEEE Trans. Netw. Service Manag.*, vol. 21, no. 1, pp. 1190–1212, Feb. 2024.
- [2] O. Urečková, M. Jureček, M. Stamp, F. Di Troia, and R. Lórencz, "Classification and online clustering of zero-day malware," *J. Comput. Virol. Hacking Tech.*, vol. 20, pp. 579–592, Feb. 2024.
- [3] P. Verma, N. Bharot, J. G. Breslin, D. O'Shea, A. Vidyarthi, and D. Gupta, "Zero-day guardian: A dual model enabled federated learning framework for handling zero-day attacks in 5G enabled IIoT," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 3856–3866, Feb. 2024.
- [4] R. Ahmad, I. Alsmadi, W. Alhamdani, and L. Tawalbeh, "A deep learning ensemble approach to detecting unknown network attacks," *J. Inf. Secur. Appl.*, vol. 67, Jun. 2022, Art. no. 103196.
- [5] R. Yoshihashi, W. Shao, R. Kawakami, S. You, M. Iida, and T. Naemura, "Classification-reconstruction learning for open-set recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 4011–4020.
- [6] A. A. Korba, A. Boualouache, and Y. Ghamri-Doudane, "Zero-X: A blockchain-enabled open-set federated learning framework for zero-day attack detection in IoV," *IEEE Trans. Veh. Technol.*, vol. 73, no. 9, pp. 12399–12414, Sep. 2024.
- [7] C. S. Shieh, F. A. Ho, M. F. Horng, T. T. Nguyen, and P. Chakrabarti, "Open-set recognition in unknown DDoS attacks detection with reciprocal points learning," *IEEE Access*, vol. 12, pp. 56461–56476, 2024.
- [8] M. Soltani, B. Ousat, M. J. Siavoshani, and A. H. Jahangir, "An adaptable deep learning-based intrusion detection system to zero-day attacks," *J. Inf. Secur. Appl.*, vol. 76, Aug. 2023, Art. no. 103516.
- [9] Y. Wu, Y. Hu, J. Wang, M. Feng, A. Dong, and Y. Yang, "An active learning framework using deep Q-network for zero-day attack detection," *Comput. Secur.*, vol. 139, Apr. 2024, Art. no. 103713.
- [10] M. Verkerken et al., "A novel multi-stage approach for hierarchical intrusion detection," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 3, pp. 3915–3929, Sep. 2023.
- [11] T. Das, O. A. Hamdan, R. M. Shukla, S. Sengupta, and E. Arslan, "UNR-ID: Intrusion detection dataset using network port statistics," in *Proc. IEEE 20th Consum. Commun. Netw. Conf. (CCNC)*, 2023, pp. 497–500.
- [12] S. Samarakoon et al., 2022, "5G-NIDD: A comprehensive network intrusion detection dataset generated over 5G wireless network," Dataset, IEEE Dataport.
- [13] M. Zolanvari, M. A. Teixeira, L. Gupta, K. M. Khan, and R. Jain, (Washington Univ. St. Louis, St. Louis, MO, USA). *WUSTL-IIOT-2021 Dataset for IIoT Cybersecurity Research*. (2021). [Online]. Available: <http://www.cse.wustl.edu/jain/iiot2/index.html>.
- [14] L. Yang, A. Moubayed, and A. Shami, "MTH-IDS: A multitiered hybrid intrusion detection system for Internet of Vehicles," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 616–632, Jan. 2022.
- [15] X.-H. Nguyen and K.-H. Le, "Robust detection of unknown DoS/DDoS attacks in IoT networks using a hybrid learning model," *Internet Things*, vol. 23, Oct. 2023, Art. no. 100851.
- [16] S. Watanabe, "Tree-structured Parzen estimator: Understanding its algorithm components and their roles for better empirical performance," 2023, *arXiv:2304.11127*.
- [17] R. Garnett, *Bayesian Optimization*. Cambridge, U.K.: Cambridge Univ. Press, 2023.
- [18] S. Theodoridis, *Machine Learning: A Bayesian and Optimization Perspective*. Cambridge, MA, USA: Academic, 2015.
- [19] M. J. Hashemi, E. Keller, and S. Tizpaz-Niari, "Detecting unseen anomalies in network systems by leveraging neural networks," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 3, pp. 2515–2528, Sep. 2023.
- [20] A. H. Celdrán et al., "RL and fingerprinting to select moving target defense mechanisms for zero-day attacks in IoT," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 5520–5529, 2024.
- [21] Y. Cao, K. Liu, Y. Lin, L. Wang, and Y. Xia, "Deep-reinforcement-learning-based self-evolving moving target defense approach against unknown attacks," *IEEE Internet Things J.*, vol. 11, no. 20, pp. 33027–33039, Oct. 2024.
- [22] A. Touré, Y. Imine, A. Semnont, T. Delot, and A. Gallais, "A framework for detecting zero-day exploits in network flows," *Comput. Netw.*, vol. 248, Jun. 2024, Art. no. 110476.
- [23] A. Bendale and T. E. Boult, "Towards open set deep networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 1563–1572.
- [24] U. Zukaib, X. Cui, C. Zheng, D. Liang, and S. U. Din, "Meta-fed IDS: Meta-learning and federated learning based fog-cloud approach to detect known and zero-day cyber attacks in IIoT networks," *J. Parallel Distrib. Comput.*, vol. 192, Oct. 2024, Art. no. 104934.
- [25] S. I. Popoola, R. Ande, B. Adebisi, G. Gui, M. Hammoudeh, and O. Jogunola, "Federated deep learning for zero-day botnet attack detection in IoT-edge devices," *IEEE Internet Things J.*, vol. 9, no. 5, pp. 3930–3944, Mar. 2022.
- [26] C. Kim, S.-Y. Chang, J. Kim, D. Lee, and J. Kim, "Automated, reliable zero-day malware detection based on autoencoding architecture," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 3, pp. 3900–3914, Sep. 2023.
- [27] T. Li, Z. Hong, W. Feng, L. Yu, and Z. Wen, "MS-ZeroWall: Detecting zero-day multi-step attack in smart home using VAE and HMM," *IEEE Trans. Veh. Technol.*, vol. 73, no. 9, pp. 13278–13291, Sep. 2024.
- [28] I. Mbona and J. H. P. Eloff, "Detecting zero-day intrusion attacks using semi-supervised machine learning approaches," *IEEE Access*, vol. 10, pp. 69822–69838, 2022.
- [29] T. Zoppi, A. Ceccarelli, and A. Bondavalli, "Unsupervised algorithms to detect zero-day attacks: strategy and application," *IEEE Access*, vol. 9, pp. 90603–90615, 2021.
- [30] B. I. Hairab, M. S. Elsayed, A. D. Jurcut, and M. A. Azer, "Anomaly detection based on CNN and Regularization techniques against zero-day attacks in IIoT networks," *IEEE Access*, vol. 10, pp. 98427–98440, 2022.
- [31] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. 4th Int. Conf. Inf. Syst. Security Privacy (ICISSP)*, 2018, pp. 108–116.
- [32] V. Böhm and U. Seljak, "Probabilistic autoencoder," 2022, *arXiv:2006.05479*.
- [33] F. T. Liu, K. M. Ting, and Z. H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Min.*, 2008, pp. 413–422.
- [34] T. R. Bandaragoda, K. M. Ting, D. Albrecht, F. T. Liu, Y. Zhu, and J. R. Wells, "Isolation-based anomaly detection using nearest-neighbor ensembles," *Comput. Intell.*, vol. 34, pp. 968–998, Nov. 2018.
- [35] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation-based anomaly detection," *ACM Trans. Knowl. Disc. Data*, vol. 6, no. 1, p. 3, 2012.
- [36] M. M. Breunig, H. P. Kriegel, R. T. Ng, and J. Sander, "LOF: Identifying density-based local outliers," *ACM SIGMOD Rec.*, vol. 29, no. 2, pp. 93–104, 2000.
- [37] L. Shu, H. Xu, and B. Liu, "DOC: Deep open classification of text documents," in *Proc. 2017 Conf. Empir. Methods Nat. Lang. Process.*, 2017, pp. 2911–2916.
- [38] S. Shen, C. Cai, Z. Li, Y. Shen, G. Wu, and S. Yu, "Deep Q-network-based heuristic intrusion detection against edge-based IIoT zero-day attacks," *Appl. Soft Comput.*, vol. 150, Jan. 2024, Art. no. 111080.
- [39] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2019, pp. 2623–2631.
- [40] C. K. Williams and C. E. Rasmussen, *Gaussian Processes for Machine Learning*. Cambridge, MA, USA: MIT Press, 2006.



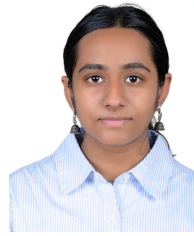
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